

Forms, validation, and effects

Module 3 - Lesson 2

Overview

Forms are how users give your app data. Controlled inputs put React in charge of form state, validation catches bad input early, and effects let components synchronize with the outside world.

Key Concepts

- A controlled input derives its value from state and updates that state on change.
- Validation can run as the user types, on blur, or on submit; show errors near the field.
- `useEffect` runs side effects (fetching, subscriptions, timers) after render.
- The dependency array controls when an effect re-runs; empty means mount-only, omitted means every render.
- Always clean up effects that create subscriptions or timers to avoid leaks.
- Use `useCallback`/`useMemo` sparingly to keep referential stability where it matters.

Example

```
useEffect(() => {  
  const timer = setInterval(() => setNow(new Date()), 1000);  
  return () => clearInterval(timer); // cleanup  
}, []);
```

Summary

Controlled forms keep validation close to the input, and effects handle the external world without touching render. Cleanup functions prevent stale timers and subscriptions.

Practice Checklist

- Build a controlled form with live validation and inline error messages.
- Fetch data on mount using `useEffect` and handle the loading state.
- Write an effect with a cleanup function and confirm no warnings in the console.